NAME:__________________________ PDG:________ Register No: ____

ANDERSON JUNIOR COLLEGE

Preliminary Examinations 2011

CHEMISTRY **9647/02**

Higher 2 **19 September 2011**

Paper 2 Structured Questions **2 hours**

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, PDG and register number on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **ALL** questions.
A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
5	
Total	

This document consists of **17** printed pages.

1 Planning (P)

You are given two different acids to investigate how the enthalpy change of neutralisation, ΔH_{neut}, varies when the acids are neutralised with aqueous sodium hydroxide, NaOH.

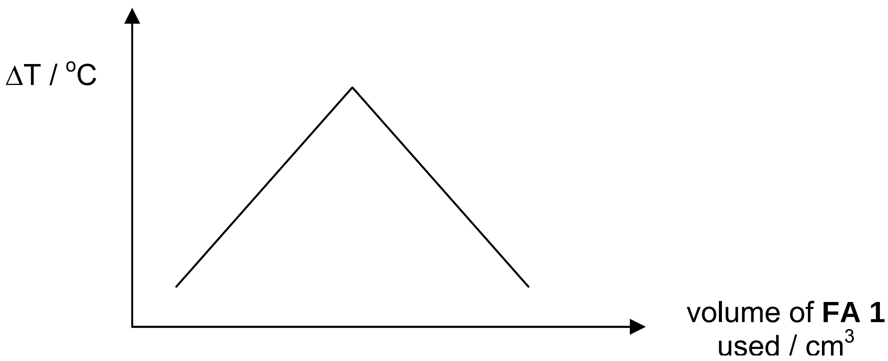
FA 1 is a solution of sulfuric acid
FA 2 is a solution of hydrochloric acid
FA 3 is 1.00 mol dm^{-3} sodium hydroxide

You may use the following information in answering the question.

The enthalpy change of neutralisation is the enthalpy change when 1 mol of water, H_2O, is formed in the neutralisation of an acid and base.

$$H^+(aq) + OH^-(aq) \longrightarrow H_2O(l)$$

You are to design a series of experiments using different volumes of **FA 1** and **FA 3** which together give a total volume of 50 cm^3. The temperature change, ΔT, for each experiment will be determined and can be used to plot a graph of ΔT against volume of **FA 1** used.



Data from the graph can be used to determine the concentration of sulfuric acid in **FA 1**, and the enthalpy change of neutralisation, ΔH_{neut}.

(a) State and explain the minimum number of experiments you will need to perform to plot the graph shown above.

....

....
[1]

(b) **(i)** Suggest a reason why the total volume is kept constant at 50 cm^3.

....

....

(ii) State one consideration to be taken when deciding the minimum volume of **FA 1** to be used.

....

....
[2]

The following procedure was carried out.

Step 1 Transfer 10 cm^3 of **FA 1** into a cup labelled **FA 1**.
Step 2 Transfer 40 cm^3 of **FA 3** into another cup labelled **FA 3**.
Step 3 Stir and measure the temperature of the **FA 1** solution.
Step 4 Add the contents of the **FA 3** cup to the **FA 1** cup. Use the thermometer to stir the mixture and measure the maximum temperature of the mixture.
Step 5 Wash and dry both the cups.
Step 6 Repeat steps 1 to 5 above for as many times as you have stated in **(a)**.

(c) Suggest the most appropriate apparatus for each of the following:

(i) to contain **FA 1** / **FA 3** ……………………………………………………

(ii) to transfer **FA 1** / **FA 3** ……………………………………………………

[1]

(d) Indicate, on the graph given on page 2, the data that you would use to determine the concentration of sulfuric acid in **FA 1**, and the enthalpy change of neutralisation, ΔH_{neut}.

[2]

(e) Using the data stated in **(d)**, show the mathematical expression for

(i) the concentration of sulfuric acid in **FA 1**.

(ii) the enthalpy change of neutralisation of sodium hydroxide with sulfuric acid.

[4.2 J of heat energy raise the temperature of 1 cm^3 of any solution by 1 ^oC]

[2]

(f) You are given two unlabelled bottles containing 1 mol dm^{-3} sulfuric acid and 1 mol dm^{-3} hydrochloric acid.

Outline the steps you will take to determine which bottle contains hydrochloric acid using only the apparatus and chemicals you are given above.

Hence explain how your proposed steps will allow the identity of hydrochloric acid to be determined.

....

....

....

....

....

....

....

....

....

....

....

....

....

....

....

....

....

....

....

[4]

[Total: 12]

2 **(a)** **(i)** State two assumptions of the kinetic theory of gases.

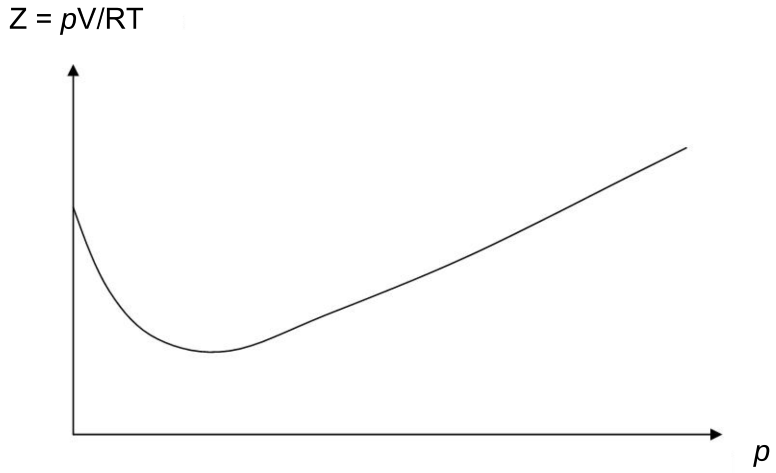
…………………………………………………………………………………………………...

…………………………………………………………………………………………………...

…………………………………………………………………………………………………...

…………………………………………………………………………………………………...

(ii) Carbon dioxide is a non–ideal gas. The graph below shows experimental values of compressibility factor (Z) plot for one mole of carbon dioxide gas. (Z = *p*V/RT)



On the plot given, sketch the graph to illustrate the behavior of an ideal gas.

(iii) State the type of hybridisation present for the carbon atom in carbon dioxide molecule.

…………………………………………………………………………………………………...

[4]

(b) In the industrial liquefaction of air, high–pressure air is pre–cooled in a coil surrounded by cold water. It is then allowed to expand into a region of low pressure, whereupon it cools down by a large amount. An ideal gas does not show this behaviour.

(i) Suggest why the expansion of air into a region of low pressure is an endothermic process.

…………………………………………………………………………………………………...

…………………………………………………………………………………………………...

…………………………………………………………………………………………………...

(ii) Real gases like carbon dioxide can be liquefied at room temperature just by pressuring them. Explain why the application of pressure causes the gas to liquefy.

…………………………………………………………………………………………………...

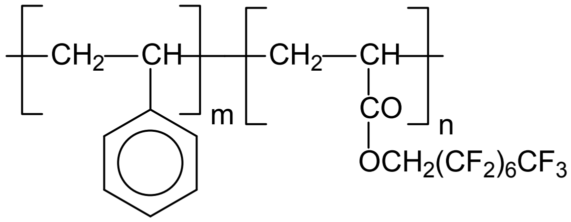
…………………………………………………………………………………………………...

…………………………………………………………………………………………………...

[3]

(c) Gases can only be liquefied by pressure alone if their temperature is below their critical temperature. Beyond the critical temperature and pressure, the compound exists as supercritical fluid, a state that resembles a gas but has density closer to the liquid phase.

Supercritical carbon dioxide has been widely used as a solvent for dry cleaning. One way of overcoming the problem of poor solubility of oils in carbon dioxide involves the use of surfactants. An example of such surfactant polymer is shown below.



DeSimones' surfactant polymer

(i) By reference to the type of intermolecular forces, explain how the use of surfactants like DeSimones' surfactant polymer helps supercritical carbon dioxide become a more effective dry cleaning solvent.

…………………………………………………………………………………………………...

…………………………………………………………………………………………………...

…………………………………………………………………………………………………...

…………………………………………………………………………………………………...

…………………………………………………………………………………………………...

(ii) Ammonia is another real gas. Explain how the critical temperature of ammonia would compare to that of carbon dioxide.

…………………………………………………………………………………………………...

…………………………………………………………………………………………………...

…………………………………………………………………………………………………...

[3]

[Total: 10]

3 **(a)** On heating gaseous pentachloride, the following equilibrium is set up:

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

(i) Use the Data Booklet to calculate the relative molecular masses (M_r) of PCl_5, PCl_3 and Cl_2.

M_r of PCl_5 ....

M_r of PCl_3 ....

M_r of Cl_2 ....

When 15.0 g of phosphorus pentachloride was put into a sealed evacuated vessel of capacity 1 L, and heated to 473 K, the pressure increased to 3.10×10^5 Pa.

(ii) Use the above data and the ideal gas equation to calculate the average M_r of the gaseous mixture.

(iii) From your answers in **(i)** and **(ii)**, use the two formulae below to calculate a value for x, and to show that K_c under the above conditions has a numerical value of 1.0×10^{-2}.

fraction dissociated (x) = $\dfrac{M_r(PCl_5) - \text{average } M_r}{\text{average } M_r}$

and $K_c = \dfrac{x^2}{1-x}$

(iv) $\Delta G^{\ominus}$ and K_c are related by the following equation

$$\Delta G^{\ominus} = -2.303 \text{ RT lg } K_c$$

where $\Delta G^{\ominus}$ is the standard free energy change **in J mol^{-1}**, R is the gas constant and T is the temperature at which the equilibrium is established.

Calculate the standard free energy change of the above equilibrium.

(v) Calculate the enthalpy change of the above equilibrium using the following enthalpy changes of formation.

[ΔH_f (PCl_5) = –443 kJ mol^{-1}; ΔH_f (PCl_3) = –319 kJ mol^{-1}]

(vi) Hence calculate the entropy change of the above equilibrium.

[10]

(b) Titanium is strong, of low density and resistant to chemical attack. Although it is expensive to produce, it is highly in demand in the aerospace industry and manufacture of implants in medicine.

Extraction of titanium from its ore cannot be carried out using carbon as a reducing agent. Given that $\Delta H^\ominus$ and $\Delta S^\ominus$ for the reduction of titanium dioxide to titanium using carbon is +720 kJ mol^{-1} and +365 J mol^{-1} K^{-1} respectively, calculate the temperature at which the reaction is feasible.

Hence, suggest a reason why this method of reduction is not carried out in the industry.

……....

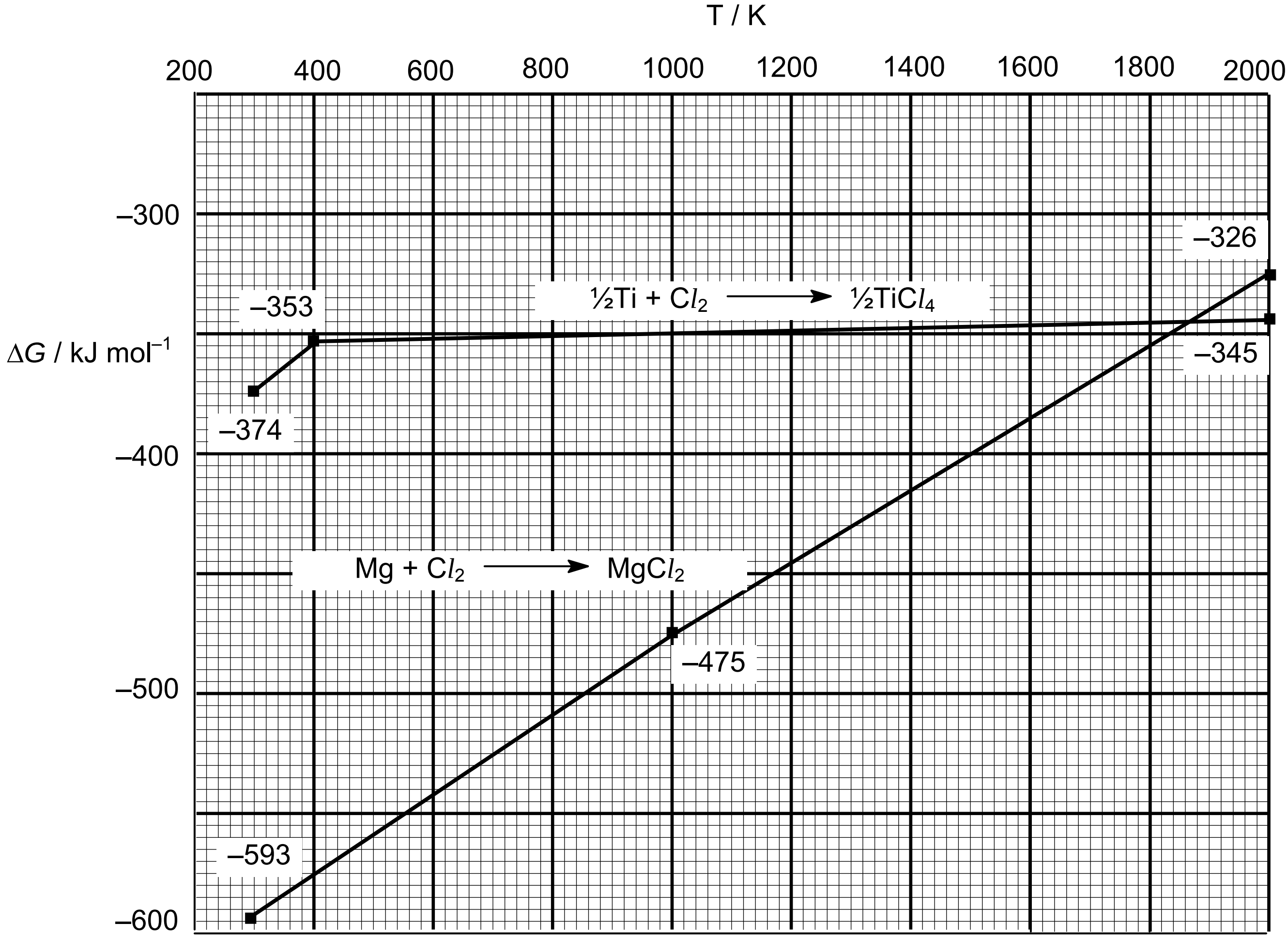
……....

[2]

(c) At present, titanium is more commonly produced by reducing titanium tetrachloride with magnesium. Titanium tetrachloride is a colourless liquid that boils at 400 K.
To optimise this reduction process, it is necessary to know how ΔG of the reaction changes with temperature. Graphs of ΔG against T are known as Ellingham diagrams. The Ellingham diagram below is for the following reactions.

reaction 1: $\frac{1}{2}Ti + Cl_2 \longrightarrow \frac{1}{2}TiCl_4$

reaction 2: $Mg + Cl_2 \longrightarrow MgCl_2$



(i) Explain, with the aid of equations, why the graph for reaction 1 becomes more positive between 300 K and 400 K but becomes almost independent of temperature from 400 K to 2000 K.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

Industrially, the reduction of titanium tetrachloride by magnesium is carried out at 1100 K in an atmosphere of argon.

(ii) Suggest a reason for the reduction to be carried out in an atmosphere of argon.

…………………………………………………………………………………………………...

…………………………………………………………………………………………………...

(iii) Write a balanced equation for the reaction at this temperature.

…………………………………………………………………………………………………...

Hence, by reference to the diagram,

(iv) evaluate ΔG at 1100 K.

(v) state the temperature above which it is **not** feasible to reduce $TiCl_4$ with Mg.

…………………………………………………………………………………………………...

[8]

[Total: 20]

4 **(a)** The Tollen's reagent is commonly used in qualitative analysis of unknown organic compounds. One method of preparing the reagent involves addition of aqueous ammonia directly to silver nitrate solution in a continuous manner.
With the addition of some ammonia solution to silver nitrate solution, silver(I) oxide will be formed and precipitates out as a brown solid. As more ammonia solution is added, the precipitate dissolves to give a clear solution. This resultant solution is the Tollen's reagent and it contains a colourless complex **E**.

(i) Explain, with the aid of equations, how the brown solid is formed.

……………………………………………………………………………………………...

……………………………………………………………………………………………...

……………………………………………………………………………………………...

(ii) Suggest the formula of silver complex **E** present in the resultant Tollen's reagent.

……………………………………………………………………………………………...

(iii) Write the electronic configuration of silver in complex **E** in Tollen's reagent.

……………………………………………………………………………………………...

(iv) Hence, explain why complex **E** is colourless.

……………………………………………………………………………………………...

……………………………………………………………………………………………...

(v) The Tollen's reagent may be used to apply a silver mirror to glassware, e.g. the inside of an insulated vacuum flask.

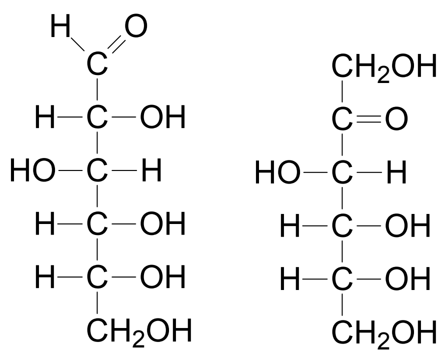
Suggest how this can be carried out in the laboratory.

……………………………………………………………………………………………...

……………………………………………………………………………………………...

[7]

(b) Besides Tollen's reagent, Fehling's solution can also be used to distinguish between different types of monosaccharides, which consists of polyhydroxylated aldehydes (aldoses) and ketones (ketoses) such as glucose and fructose respectively.



glucose fructose

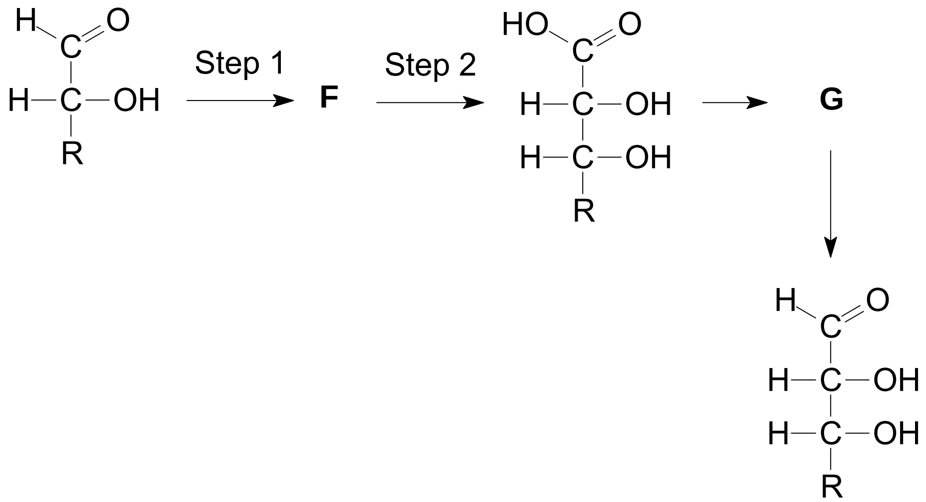
(i) Describe what you might observe when both compounds above are reacted with Fehling's solution. Draw the structure of any organic product(s) that would be formed during this test.

……………………………………………………………………………………………………...

……………………………………………………………………………………………………...

(ii) The Kiliani–Fischer synthesis lengthens the aldose chain by one carbon atom and can be used to unravel the stereochemical relationships among simple carbohydrates.

The synthesis is carried out using the following steps.



Suggest the reagents and conditions used in steps 1 and 2. Give the structural formulae of the intermediates **F** and **G**.

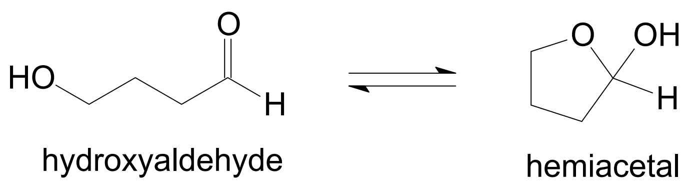
Reagents and conditions:

Step 1	
Step 2	

Intermediate **F**	Intermediate **G**

(iii) Draw the structure of the product formed when glucose is subjected to the Kiliani–Fischer synthesis.

(iv) Glucose may be converted to the cyclic form which is known as a hemiacetal. The reaction may be represented using a simpler hydroxyaldehyde as follows:



State the type of reaction for the forward reaction.

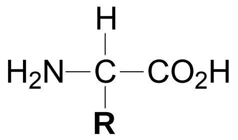
Type of reaction: ……………………………………………………………………………………

[8]

[Total: 15]

5 Ovalbumin is the predominant protein in egg white, comprising approximately 54% of the total protein. It is one of only two pure proteins that can adequately meet nutritional requirements for all amino acids, including the essential amino acids. Some of the amino acids found in ovalbumin are as follows:

Basic Structure of α–amino acid



amino acids	**formula of –R group**
alanine (ala)	$–CH_3$
aspartic acid (asp)	$–CH_2CO_2H$
glycine (gly)	–H
lysine (lys)	$–(CH_2)_4NH_2$
serine (ser)	$–CH_2OH$

(a) A polypeptide containing eight peptide linkages was extracted by hydrolysis of ovalbumin using biological proteases. Partial hydrolysis of this polypeptide produces smaller tetrapeptides, as follows:

ala–ala–ser–asp

ala–ser–asp–ala

asp–ala–ala–lys

ala–lys–gly–lys

(i) Deduce the order in which the amino acids are bonded together in the polypeptide, explaining your reasoning.

…………………………………………………………………………………………………...

…………………………………………………………………………………………………...

(ii) Draw the displayed formula of the structure present when the tetrapeptide, ala–lys–gly–lys is in a solution of pH 1.0.

[4]

(b) When food is cooked, some of its proteins become denatured. An example of the denaturation process is observed when ovalbumin is boiled in water where the soluble globular protein gets converted into insoluble fibrous proteins.

(i) What is meant by the term *denaturation*?

…………………………………………………………………………………………………...

…………………………………………………………………………………………………...

(ii) Fibrous proteins have high *β–pleated* sheet content. Draw a diagram showing the possible structure of the insoluble fibrous protein formed when ovalbumin is boiled.

(iii) The soluble globular form is formed by the folding of polypeptide chains due to the interactions of the R groups.
Describe the possible R group interactions present in ovalbumin, illustrating your answer with suitable pairs of amino acids from the table above.

…………………………………………………………………………………………………...

…………………………………………………………………………………………………...

…………………………………………………………………………………………………...

…………………………………………………………………………………………………...

…………………………………………………………………………………………………...

…………………………………………………………………………………………………...

…………………………………………………………………………………………………...

…………………………………………………………………………………………………...

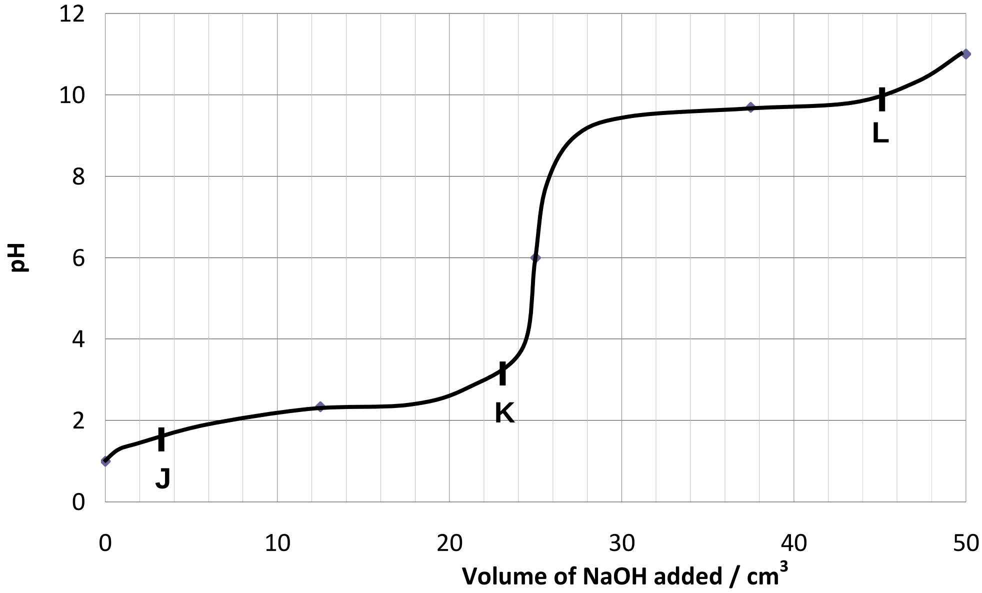
…………………………………………………………………………………………………...

[7]

(c) **(i)** Write an equation to show how the zwitterionic form of alanine reacts with OH^-.

…………………………………………………………………………………………………...

(ii) The pH change when 1 mol dm^{-3} of NaOH is added to 50 cm^3 of a 1 mol dm^{-3} solution of alanine in a solution of pH 1 is shown below.



There are two pK_a values associated with alanine: 2.3 and 9.7.

Make use these pK_a values to suggest the major organic species present at point **J, K** and **L**.

J	**K**
L	

(iii) Hence deduce the isoelectric point of alanine.

……………………………………………………………………………………………………..

[4]
[Total: 15]